Trigonometry: Reciprocal Trigonometric Functions

Questions 1–39

Question 1. Sketch the following graphs for $0° < x < 360°$:

a) $y = \operatorname{cosec} x$
b) $y = \sec x$
c) $y = \cot x$

Question 2. Sketch the following graphs for $0 < x < 2\pi$:

a) $y = \operatorname{cosec} x$
b) $y = \sec x$
c) $y = \cot x$

Question 3. Sketch the following graphs for $0° < x < 360°$:

a) $y = 2\operatorname{cosec} x$
b) $y = \frac{1}{3}\sec x$
c) $y = -\cot x$
d) $y = \operatorname{cosec}(x - 90°)$
e) $y = \sec(x + 60°)$
f) $y = \cot(x - 270°)$

Question 4. Sketch the following graphs for $0 < x < 2\pi$:

a) $y = \frac{1}{4}\operatorname{cosec} x$
b) $y = \frac{4}{3}\sec x$
c) $y = \cot(-x)$
d) $y = \operatorname{cosec}(x + \frac{\pi}{2})$
e) $y = \sec(x - \frac{\pi}{3})$
f) $y = \cot(x + \frac{2\pi}{3})$

Question 5. Prerequisite knowledge for Q5–Q6: Combinations of Graph Transformations Sketch the following graphs for $0° < x < 360°$:

a) $y = 1 + 3\operatorname{cosec} x$
b) $y = 2 - \sec x$
c) $y = -\cot(x - 30°)$
d) $y = \operatorname{cosec}(2x - 90°)$
e) $y = \sec(2x + 60°)$
f) $y = \cot(\frac{1}{2}x - 30°)$

Question 6. Sketch the following graphs for $0 < x < 2\pi$:

a) $y = \frac{1}{2}\operatorname{cosec} x - 1$
b) $y = \frac{1}{3}\sec x - 2$
c) $y = \cot(-2x)$
d) $y = \operatorname{cosec}(\frac{1}{2}x + \frac{\pi}{4})$
e) $y = \sec(\frac{1}{2}x - \frac{\pi}{3})$
f) $y = \cot(2x + \frac{\pi}{3})$

Question 7. Solve the following equations:

a) $\operatorname{cosec}\theta = 2,\ 0° < \theta < 360°$

b) $\sec\theta = 3,\ 0 < \theta < 2\pi$

c) $\cot\theta = 4,\ 0° < \theta < 360°$

d) $\operatorname{cosec}\theta = -\frac{5}{4},\ 0 < \theta < 2\pi$

e) $\sec\theta = -\frac{9}{2},\ 0° < \theta < 360°$

f) $\cot\theta = -\frac{18}{5},\ 0 < \theta < 2\pi$

Question 8. State the values of x for which $y = \operatorname{cosec} x$ is an increasing function, for $0 < x < 2\pi$

Question 9. State the values of x for which $y = \sec x$ is a decreasing function, for $0° < x < 360°$

Question 10. It is given that $\operatorname{cosec} x = \frac{7}{2}$ and x is acute. Find the exact values of:

a) $\sin x$
b) $\cos x$
c) $\tan x$
d) $\sec x$
e) $\cot x$

Question 11. It is given that $\sec x = \frac{9}{4}$ and x is acute. Find the exact values of:

a) $\cos x$
b) $\sin x$
c) $\tan x$
d) $\operatorname{cosec} x$
e) $\cot x$

Question 12. It is given that $\cot x = \frac{12}{11}$ and x is acute. Find the exact values of:

a) $\tan x$
b) $\sin x$
c) $\cos x$
d) $\operatorname{cosec} x$
e) $\sec x$

Question 13. It is given that $\operatorname{cosec} x = \frac{10}{9}$ and x is obtuse. Find the exact values of:

a) $\sin x$
b) $\cos x$
c) $\tan x$
d) $\sec x$
e) $\cot x$

Question 14. It is given that $\sec x = -\frac{12}{5}$ and $180° < x < 270°$. Find the exact values of:

a) $\cos x$
b) $\sin x$
c) $\tan x$
d) $\operatorname{cosec} x$
e) $\cot x$

Question 15. It is given that $\cot x = -\frac{2}{7}$ and $270° < x < 360°$. Find the exact values of:

a) $\tan x$
b) $\sin x$
c) $\cos x$
d) $\operatorname{cosec} x$
e) $\sec x$

Question 16. It is given that $8\sec^2 x - 14\sec x + 3 = 0$ and x is acute. Find the exact values of:

a) $\sec x$
b) $\cos x$
c) $\sin x$
d) $\tan x$
e) $\operatorname{cosec} x$
f) $\cot x$

Question 17. Without using any calculus, state the maximum value of the curve with equation $y = \frac{2}{\operatorname{cosec}^2 x + 2}$

Question 18. Write the following trigonometric expressions in terms of $\tan x$, $\operatorname{cosec} x$, $\sec x$ and $\cot x$ only:

a) $\frac{\sin x}{\cos^2 x}$

b) $\frac{\cos x}{\sin^2 x}$

Question 19. Simplify the following trigonometric expressions as far as possible:

a) $\operatorname{cosec} x \sin x$

b) $\sec x \cos x$

c) $\cot x \tan x$

Question 20. Simplify the following trigonometric expressions as far as possible, writing your answers in terms of $\sin x$ and $\cos x$ only:

a) $\operatorname{cosec} x \sec x$

b) $\operatorname{cosec} x \cot x$

c) $\sec x \cot x$

d) $\operatorname{cosec} x \sec x \cot x$

Question 21. Simplify the following trigonometric expressions as far as possible:

a) $\frac{\operatorname{cosec} x}{\sec x}$
b) $\frac{\operatorname{cosec} x}{\cot x}$
c) $\frac{\sec x}{\operatorname{cosec} x}$
d) $\frac{\sec x}{\cot x}$
e) $\frac{\cot^2 x}{\sec x}$
f) $\frac{\cot^2 x \sin^3 x}{\operatorname{cosec} x \sec x}$

Question 22. Solve the following equations:

a) $\operatorname{cosec}\theta = \sec\theta,\ 0° < \theta < 360°$
b) $\cot\theta = -\sec\theta,\ 0 < \theta < 2\pi$
c) $\sec\theta = \cot\theta,\ 0° < \theta < 360°$
d) $\cot\theta = \tan\theta,\ 0 < \theta < 2\pi$

Question 23.

a) Sketch the graphs of $y = \operatorname{cosec} x$ and $y = 2\sec x$ on the same axes in radians between 0 and 2π

b) Determine the coordinates of where the two graphs intersect.

Question 24. State the values of k for which $\sec x - 3 + k = 0$ has real solutions.

Question 25. Starting from $\sin^2 x + \cos^2 x \equiv 1$

a) derive an identity for $\cot x$ and $\operatorname{cosec} x$,

b) derive an identity for $\tan x$ and $\sec x$

Question 26. Simplify the following trigonometric expressions:

a) $\frac{1+\cot^2 x}{\operatorname{cosec}^2 x}$

b) $\frac{1+\tan^2 x}{\sec^2 x}$

c) $\frac{1+\tan^2 x}{1+\cot^2 x}$

d) $\frac{2+2\cot^2 x}{5+5\tan^2 x}$

Question 27. Solve the following equations:

a) $\cot^2\theta + 3\operatorname{cosec}\theta - 3 = 0,\ 0° < \theta < 360°$

b) $\tan^2\theta - \sec\theta - 11 = 0,\ 0 < \theta < 2\pi$

c) $\operatorname{cosec}^2\theta + 5\cot\theta - 15 = 0,\ 0° < \theta < 360°$

d) $\sec^2\theta + 6\tan\theta - 28 = 0,\ 0 < \theta < 2\pi$

Question 28. Solve the following equations:

a) $3\operatorname{cosec}^2\theta + 2\cot\theta - 4 = 0,\ 0° < \theta < 360°$

b) $4\sec^2\theta - 4\tan\theta - 19 = 0,\ 0 < \theta < 2\pi$

c) $6\cot^2\theta + 23\operatorname{cosec}\theta + 13 = 0,\ 0° < \theta < 360°$

d) $18\tan^2\theta + 9\sec\theta - 2 = 0,\ 0 < \theta < 2\pi$

Question 29. Show that $3\tan^2 x + 4k\sec x + \frac{4k^2}{3} + 3$ can be written in the form $3\left(\sec x + \frac{2k}{3}\right)^2$

Question 30. It is given that $10\tan^2 x + 17\sec x + 16 = 0$ and x is obtuse. Find the exact values of:

a) $\sec x$
b) $\cos x$
c) $\sin x$
d) $\tan x$
e) $\operatorname{cosec} x$
f) $\cot x$

Question 31. It is given that $16\cot^2 x + 32\operatorname{cosec} x + 31 = 0$ and $\frac{3\pi}{2} < x < 2\pi$. Find the exact values of:

a) $\operatorname{cosec} x$
b) $\sin x$
c) $\cos x$
d) $\tan x$
e) $\sec x$
f) $\cot x$

Question 32. Prove there are no real solutions to the equation $12\cot^2\theta - 5\operatorname{cosec}\theta + 10 = 0$

Question 33. It is given that $\mathrm{p}(x) = 12x^3 + 8x^2 - 3x - 2$

a) Show that $(3x + 2)$ is a factor of $\mathrm{p}(x)$

b) Write $\mathrm{p}(x)$ as a product of three linear factors.

c) Hence prove that there are no real solutions to the equation $12\,\mathrm{cosec}^3\,\theta + 8\cot^2\theta - 3\,\mathrm{cosec}\,\theta + 6 = 0$

Question 34. Prove the following trigonometric identities:

a) $\cot x + \cos x \equiv \cot x(1 + \sin x)$

b) $\cot x + \sec x \equiv \mathrm{cosec}\,x \sec x(\sin x + \cos^2 x)$

c) $\cot x + \sin x \equiv \mathrm{cosec}\,x(\sin^2 x + \cos x)$

d) $\cot x + \tan x \equiv \mathrm{cosec}\,x \sec x$

Question 35. Prove the following trigonometric identities:

a) $\sec x - \cos x \equiv \sin x \tan x$

b) $\operatorname{cosec} x - \sin x \equiv \cos x \cot x$

c) $\tan x - \cot x \equiv \operatorname{cosec} x \sec x(2\sin^2 x - 1)$

d) $\tan x - \operatorname{cosec} x \equiv \operatorname{cosec} x(\sin x \tan x - 1)$

Question 36. Prove the following trigonometric identities:

a) $\sin^2 x - \cot^2 x \equiv \operatorname{cosec}^2 x(\sin^4 x - \cos^2 x)$

b) $\cos^2 x - \cot^2 x \equiv -\cos^2 x \cot^2 x$

c) $\sec^2 x - \operatorname{cosec}^2 x \equiv \operatorname{cosec}^2 x \sec^2 x(2\sin^2 x - 1)$

d) $\tan^2 x - \cot^2 x \equiv \operatorname{cosec}^2 x \sec^2 x(2\sin^2 x - 1)$

Question 37. Prove the following trigonometric identities:

a) $(\sec x + \cot x)^2 = \tan^2 x + 2\operatorname{cosec} x + \operatorname{cosec}^2 x$

b) $(\sec x + \operatorname{cosec} x)^2 \equiv \tan^2 x + \cot^2 x + 2(\tan x + \cot x + 1)$

c) $(\operatorname{cosec} x + \cot x)^2 \equiv 2\cot^2 x + 1 + 2\cos x(\cot^2 x + 1)$

d) $(\tan x + \sec x)^2 \equiv 2\sec^2 x(\sin x + 1) - 1$

e) $(\tan x + \sec x)^2 \equiv \frac{1+\sin x}{1-\sin x}$

Question 38.

a) Prove the following trigonometric identity:

$$\frac{\sin\theta}{1-\cos\theta} - \operatorname{cosec}\theta \equiv \cot\theta$$

b) Hence solve the equation

$$\frac{\sin\theta}{1-\cos\theta} = 10 + \cot\theta$$

for $0 < \theta < 2\pi$

Question 39. Prerequisite knowledge for Q39: Double Angle Formulae Solve the equation $\operatorname{cosec}\theta = 10\cos\theta$ for $0 < \theta < 2\pi$